

ECON 3040 - Chapter 3: Statistics Review

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What is a *statistic*?

Examples:

- ▶ 9 out of 10 dentists choose Colgate
- ▶ The richest 1% own half the wealth
- ▶ Surgeons who played more than 3 hours / week of video games had 37% fewer errors
- ▶ Vladimir Guerrero Jr. hit his first home game home run during the 150th game in the 2026 season
- ▶ The maximum mark in ECON 3040 2025 A01 was 99.6%
- ▶ The median after tax income of Canadian families was \$75,500 in 2024
- ▶ The average age of retirement in Canada is 65.4 years

How were these statistics calculated?

Statistics Review

- ▶ A **statistic** is a function of a sample of data
- ▶ An **estimator** is a statistic
- ▶ **Population parameter** $\rightarrow$ unknown
- ▶ **Estimator** $\rightarrow$ used to estimate an unknown population parameter
- ▶ The sample, y , will be considered **random**
- ▶ Since y is random, estimators using y will be **random**

Since estimators are random, they have a _____, given a special name: **sampling distribution**.

We will obtain properties of the sampling distribution to see if the estimator is “good” or not.

3.1 Random Sampling from the Population

- ▶ Typically, we want to know something about a population
- ▶ The population is considered to be very large (infinite), and contains some unknown “truth”
- ▶ We likely won’t observe the whole population, but a sample from the pop.
- ▶ We’ll use the sample, y , to estimate that something

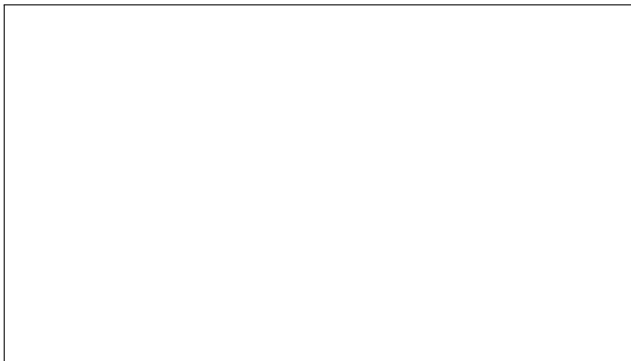
Example: Mean Height of a U of M Student

Suppose we want to know the mean height of a U of M student.

Let y = height of a student.

- ▶ Population: all students
- ▶ Population parameter of interest: μ_Y

We can't afford to observe the whole pop. We'll have to collect a sample, y .



How Should the Sample Be Selected?

We want the sample to reflect the population.

Question: How should the sample be selected from the population?

In particular we want the sample to be i.i.d.

- ▶ **Identically**
- ▶ **Independently**
- ▶ **Distributed**

The Sample Is Random

So, the sample y is random!

- ▶ Could have gotten a different y
- ▶ Parallel universe

Table 3.1: Entire population of heights (in cm). The true (unobservable) population mean and variance are $\mu_y = 176.8$ and $\sigma_y^2 = 39.7$.

177.3	170.2	187.2	178.3	170.3	179.4	181.2	180.0	173.9
178.7	171.7	160.5	183.9	175.7	175.9	182.6	181.7	180.2
181.5	176.5	162.1	180.3	175.6	174.9	165.7	172.7	178.9
175.3	178.7	175.6	166.4	173.1	173.2	175.6	183.7	181.3
174.2	180.9	179.9	171.2	171.0	178.6	181.4	175.2	182.2
171.7	178.4	168.1	186.0	189.9	173.4	168.7	180.0	175.1
175.7	180.8	176.2	170.8	177.3	163.4	186.3	177.1	191.2
171.0	180.3	169.5	167.2	178.0	172.9	176.0	176.5	171.9
175.1	184.2	165.3	180.2	178.3	183.4	173.9	178.6	177.9
184.5	184.1	180.9	187.1	179.9	167.1	172.0	167.4	172.7
171.6	186.6	182.4	185.5	174.8	178.8	192.8	179.3	172.0

From Random Samples to Estimators

How could i.i.d. be violated in the heights example?

Example: mean income of Canadians. How could i.i.d. be violated?

How should we estimate the mean height?

3.2 Estimators and Sampling Distributions

An estimator uses the sample y to “guess” something about the pop.
We collect our sample:

$$y = \{173.9, 171.7, 182.6, 181.5, 162.1, 174.9, 165.7, \\ 182.2, 171.7, 168.1, 189.9, 175.7, 163.4, 186.3, \\ 169.5, 171.9, 173.9, 172.0, 172.7, 172.0\}.$$

How should we use this sample to estimate the mean height?

3.2.1 Sample Mean

A popular choice for estimating a population mean is by using a *sample mean* (or *sample average* or just *average*)

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i \quad (3.1)$$

- ▶ From heights example: $\bar{y} = 174.1$, $\mu_y = 176.8$
- ▶ There are many ways to estimate μ_y . Examples?
- ▶ Why is (3.1) so popular?
- ▶ How good is $\bar{y}$ at estimating μ_y in general?
- ▶ To answer these questions: idea of a *sampling distribution*

A parallel universe

Recall that the sample, y , is random. Each element of y was selected randomly from the population. We could have selected a different sample of size $n = 20$. For example, in a parallel universe, we could have gotten

$$y^* = \{175.9, 175.3, 182.2, 178.6, 175.2, 180.3, 178.3, \\ 183.7, 176.0, 167.4, 178.7, 178.7, 186.0, 175.6, \\ 180.0, 168.7, 178.6, 173.1, 173.2, 187.1\},$$

where the $*$ in y^* denotes that we are in the parallel universe. In this parallel universe, we got $\bar{y}^* = 177.6$. But in every universe, the population μ_y (table 3.1) is the same.

The Sample Mean Is Random

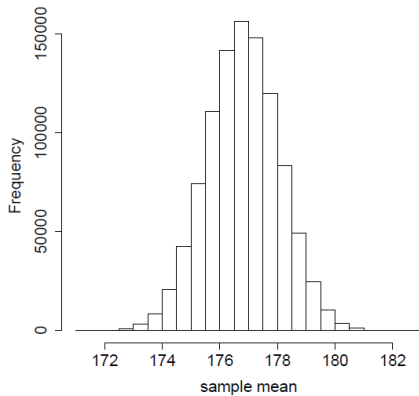
- ▶ Randomly sample from the population $\rightarrow$ get y
 - ▶ y is random
- ▶ Use y to calculate $\bar{y}$
 - ▶ $\bar{y}$ is random
 - ▶ could have gotten a different sample $\rightarrow$ could have gotten a different $\bar{y}$
 - ▶ population is always the same (μ_y)

3.2.2 Sampling Distribution of the Sample Mean

- ▶ $\bar{y}$ is a random variable (it's an estimator; all estimators are random)
- ▶ Random variables (usually) have probability functions
- ▶ $\bar{y}$ has a **sampling distribution** (probability function for an estimator)
- ▶ Sampling distribution: imagine all possible values for $\bar{y}$ that you could get — plot a histogram
- ▶ Using a computer, I drew 1 mil. different random samples of $n = 20$ from Table 3.1. Calculate $\bar{y}$ each time. Plot histogram:

Sampling Distribution of $\bar{y}$

Figure 3.1: Histogram for 1 million $\bar{y}$ s



Which Probability Function?

Which probability function is right for $\bar{y}$? Why?

- ▶ Look at Figure 3.1
- ▶ Notice the summation operator in equation (3.1)
- ▶ Answer: _____ Reason: _____

$\bar{y}$ is random. We'll derive its:

- ▶ mean
- ▶ variance

Use these to determine if it's a “good” estimator via three statistical properties:

- ▶ Bias
- ▶ Efficiency
- ▶ Consistency

3.2.3 Bias

An estimator is **unbiased** if its expected value is equal to the population parameter it's estimating.

That is, $\bar{y}$ is unbiased if

$$E[\bar{y}] = \mu_y.$$

Unbiased if it gives “the right answer on average”.

Biased if it gives the wrong answer on average.

The Sample Mean Is Unbiased

$$\begin{aligned} E[\bar{y}] &= E\left[\frac{1}{n} \sum_{i=1}^n y_i\right] \\ &= \frac{1}{n} E\left[\sum_{i=1}^n y_i\right] \\ &= \frac{1}{n} E[y_1 + y_2 + \cdots + y_n] \\ &= \frac{1}{n} (E[y_1] + E[y_2] + \cdots + E[y_n]) \\ &= \frac{1}{n} (\mu_y + \mu_y + \cdots + \mu_y) \\ &= \frac{n\mu_y}{n} = \mu_y. \end{aligned} \tag{3.2}$$

3.2.4 Efficiency

An estimator is **efficient** if it has the smallest variance among all other potential estimators (for us, potential = linear, unbiased).

Need to get the variance of $\bar{y}$.

Variance of the Sample Mean

$$\begin{aligned}\text{Var}[\bar{y}] &= \text{Var}\left[\frac{1}{n} \sum_{i=1}^n y_i\right] \\&= \frac{1}{n^2} \text{Var}\left[\sum_{i=1}^n y_i\right] \\&= \frac{1}{n^2} \text{Var}[y_1 + y_2 + \cdots + y_n] \\&= \frac{1}{n^2} (\text{Var}[y_1] + \text{Var}[y_2] + \cdots + \text{Var}[y_n]) \\&= \frac{1}{n^2} (\sigma_y^2 + \sigma_y^2 + \cdots + \sigma_y^2) \\&= \frac{n\sigma_y^2}{n^2} = \frac{\sigma_y^2}{n}.\end{aligned}\tag{3.3}$$

- ▶ Gauss–Markov theorem proves this is minimum variance
- ▶ We’ll also need this to prove consistency, and for hyp. testing

3.2.5 Consistency

Suppose we had a lot of information. ($n \rightarrow \infty$)

What value should we get for our estimator?

How would you state this mathematically?

Q) Prove that the sample mean is a consistent estimator for the population mean.

Q) Define the terms unbiasedness, efficiency, and consistency.

3.3 Hypothesis Tests (known σ_y^2)

$$\begin{aligned}H_0 : \mu_y &= \mu_{y,0} \\ H_A : \mu_y &\neq \mu_{y,0}\end{aligned}\tag{3.4}$$

- ▶ Get your random sample y
- ▶ Estimate μ_y (using $\bar{y}$ for example)
- ▶ See if $\bar{y}$ appears “close” to $\mu_{y,0}$
 - ▶ Remember, $\bar{y}$ is random! (and Normal)
- ▶ If it's close $\rightarrow$ fail to reject
- ▶ If it's far $\rightarrow$ reject

Hypothesis Test Example

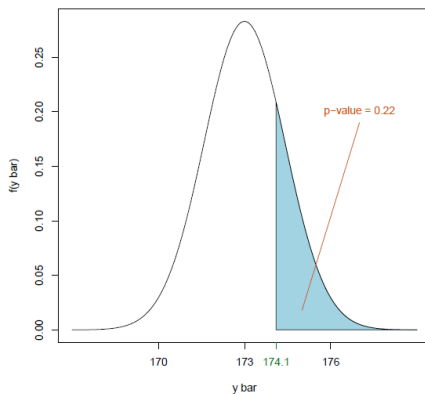
- ▶ Hypothesize that mean height of a U of M student is 173cm

$$\begin{aligned}H_0 : \mu_y &= 173 \\ H_A : \mu_y &\neq 173\end{aligned}\tag{3.5}$$

- ▶ Collect a sample: $y = \{173.9, 171.7, \dots, 172.0\}$
- ▶ Calculate $\bar{y} = 174.1$
- ▶ Suppose (very unrealistically that we know that) $\sigma_y^2 = 39.7$
- ▶ What now?

The p -value

Figure 3.2: Normal distribution with $\mu = 173$ and $\sigma^2 = 39.7/20$. Shaded area is the probability that the normal variable is greater than 174.1.



Interpreting the Test

The p -value for the above test is 0.44. How to interpret this?

3.3.1 Significance of a test

3.3.2 Type I error

3.3.3 Type II error (and power)

3.3.4 Test Statistics

- ▶ Just a more convenient way of getting the p -value for the test
- ▶ Each hypothesis test would present us with a new normal curve that we would have to draw, and calculate a new area (see Fig. 3.2)
- ▶ Instead: standardize
- ▶ This gives us one curve for all testing problems (the standard normal curve)
- ▶ Calculate a bunch of areas under the curve, and tabulate them
- ▶ Not an issue with modern computers, but this is still the way we do things
- ▶ How to get a z test statistic?
- ▶ Do a z test for our heights example.

Standard Normal Table

Table 3.2: Area under the standard normal curve, to the right of z .

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641
0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.9985
1.3	.9968	.9951	.9934	.9918	.9901	.9885	.9869	.9853	.9838	.9823
1.4	.9808	.9793	.9778	.9764	.9749	.9735	.9721	.9708	.9694	.9681
1.5	.9668	.9655	.9643	.9630	.9618	.9606	.9594	.9582	.9571	.9559
1.6	.9548	.9537	.9526	.9516	.9505	.9495	.9485	.9475	.9465	.9455
1.7	.9446	.9436	.9427	.9418	.9409	.9401	.9392	.9384	.9375	.9367
1.8	.9359	.9351	.9344	.9336	.9329	.9322	.9314	.9307	.9301	.9294
1.9	.9287	.9281	.9274	.9268	.9262	.9256	.9250	.9244	.9239	.9233
2.0	.9228	.9222	.9217	.9212	.9207	.9202	.9197	.9192	.9188	.9183
2.1	.9179	.9174	.9170	.9166	.9162	.9158	.9154	.9150	.9146	.9143
2.2	.9139	.9136	.9132	.9129	.9125	.9122	.9119	.9116	.9113	.9110
2.3	.9107	.9104	.9102	.9099	.9096	.9094	.9091	.9089	.9087	.9084
2.4	.9082	.9080	.9078	.9075	.9073	.9071	.9069	.9068	.9066	.9064
2.5	.9062	.9060	.9059	.9057	.9055	.9054	.9052	.9051	.9049	.9048
2.6	.9047	.9045	.9044	.9043	.9041	.9040	.9039	.9038	.9037	.9036
2.7	.9035	.9034	.9033	.9032	.9031	.9030	.9029	.9028	.9027	.9026
2.8	.9026	.9025	.9024	.9023	.9023	.9022	.9021	.9021	.9020	.9019
2.9	.9019	.9018	.9018	.9017	.9016	.9016	.9015	.9015	.9014	.9014
3.0	.9013	.9013	.9013	.9012	.9012	.9011	.9011	.9011	.9010	.9010
3.1	.9010	.9009	.9009	.9009	.9008	.9008	.9008	.9008	.9007	.9007
3.2	.9007	.9007	.9006	.9006	.9006	.9006	.9006	.9005	.9005	.9005
3.3	.9005	.9005	.9005	.9004	.9004	.9004	.9004	.9004	.9004	.9003
3.4	.9003	.9003	.9003	.9003	.9003	.9003	.9003	.9003	.9003	.9002

3.3.5 Critical Values

A critical value is a predetermined, maximum value for the (absolute value of the) test statistic, after which you **reject**.

Look at Table 3.2 and find that **1.96** is the 5% critical value.

So, if $|z| > 1.96$, then you would reject.

3.3.6 Confidence Intervals

What is the probability that our z statistic will be within a certain interval, if the null hypothesis is true? For example, what is the following probability?

$$\Pr(-1.96 \leq z \leq 1.96)? \quad (3.12)$$

$$\Pr\left(-1.96 \leq \frac{\bar{y} - \mu_{y,0}}{\sqrt{\sigma_y^2/n}} \leq 1.96\right) = 0.95 \quad (3.13)$$

Finally, we solve equation (3.13) so that the null hypothesis $\mu_{y,0}$ is in the middle of the probability statement:

$$\Pr\left(\bar{y} - 1.96\sqrt{\frac{\sigma_y^2}{n}} \leq \mu_{y,0} \leq \bar{y} + 1.96\sqrt{\frac{\sigma_y^2}{n}}\right) = 0.95. \quad (3.14)$$

3.4 Hypothesis Tests (unknown σ_y^2)

- ▶ Much more realistically, σ_y^2 (variance of y) will be unknown.
- ▶ Recall that:

$$\text{Var}[\bar{y}] = \frac{\sigma_y^2}{n}$$



$$z = \frac{\bar{y} - \mu_{y,0}}{\text{s.e.}(\bar{y})} = \frac{\bar{y} - \mu_{y,0}}{\sqrt{\sigma_y^2/n}}$$

- ▶ So, we need to estimate σ_y^2 in order to perform hypothesis tests.

3.4.1 Estimating σ_y^2

A “natural” estimator:

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2 \quad (3.15)$$

- ▶ Is this a good estimator? Why or why not?
- ▶ A better estimator:

$$s_y^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2 \quad (3.17)$$

- ▶ Degrees-of-freedom correction

The t Test

So:

Estimated variance of $\bar{y} = \frac{s_y^2}{n}$

We can implement hypothesis testing by replacing the unknown σ_y^2 with its estimator s_y^2 . The z test statistic now becomes:

$$\frac{\bar{y} - \mu_{y,0}}{\sqrt{s_y^2/n}} = t$$

Note: for large n , the t test is equivalent to the z test.